

DIGITAL SYSTEM DESIGN



Er. Sailesh Mishra

CHIP DESIGN ENGINEER

(Sevya Multimedia Technologies)

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI

What is Digital System Design ??

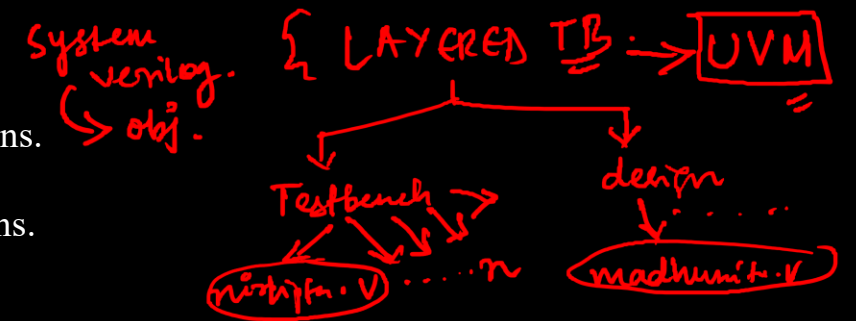
- Digital System Design is the domain of VLSI that focuses on designing, analyzing, and implementing systems whose signals and operations are represented in discrete (digital) form, typically as binary values (0s and 1s). It combines logic design, computer architecture, and circuit-level implementation to create systems that perform specific computational, control, or data-processing tasks efficiently and reliably.

What is Digital System Verification ??

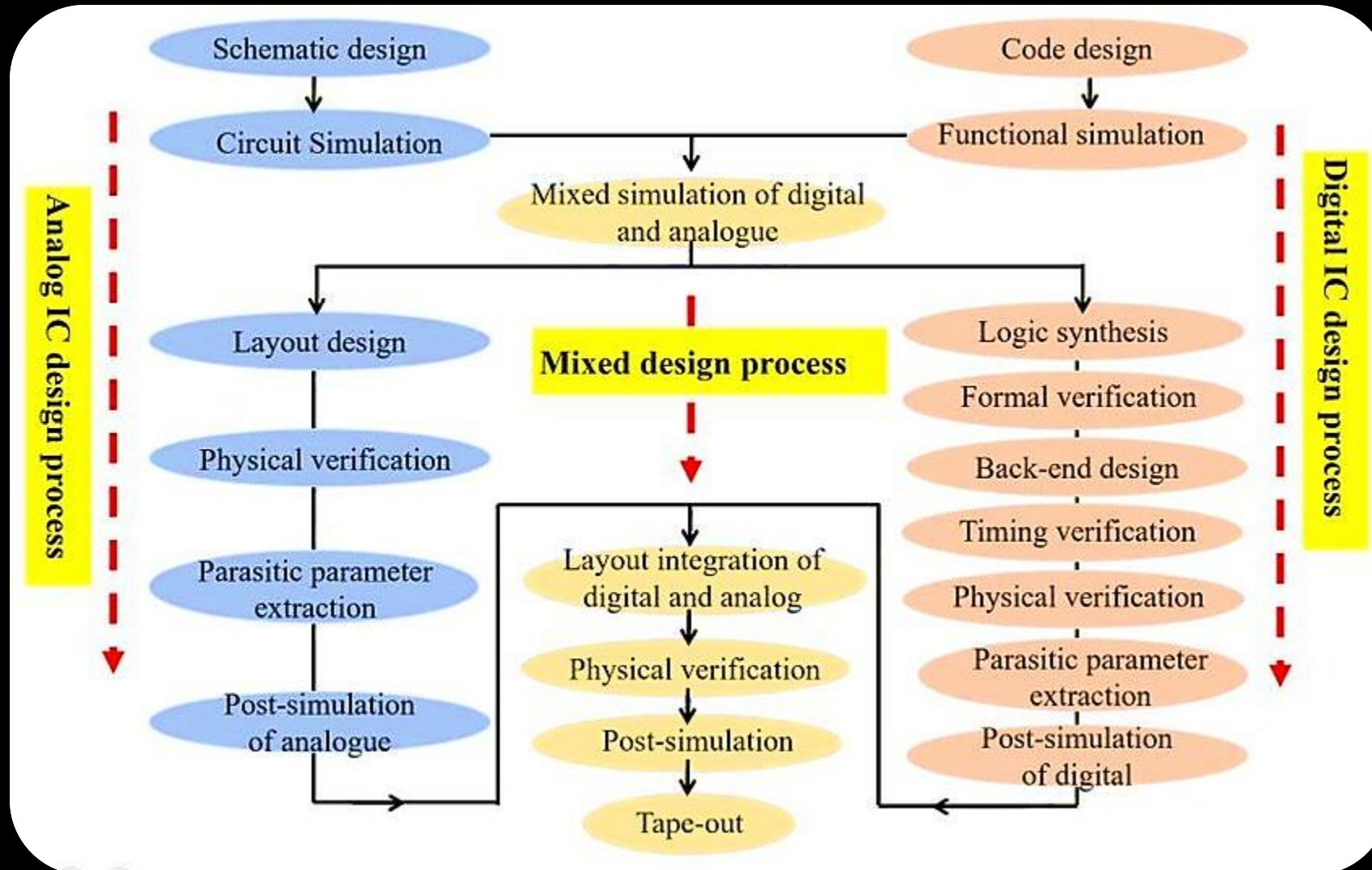
- Digital System Verification is the process of ensuring that a digital system (such as a chip, processor, or any other integrated circuit) behaves as intended before it is manufactured. It involves checking that the design meets its specifications and does not have any bugs or unexpected behavior. The process typically involves simulating the design under various conditions and inputs to verify its correctness.

- Verification is a critical step in the design process because any bugs that are missed at this stage could result in costly errors in the manufactured product. Verification is performed at different levels, including :

1. Functional Verification: Ensures that the design behaves according to its functional specifications.
2. Formal Verification: Uses mathematical methods to prove that the design meets its specifications.
3. Timing Verification: Ensures that the design meets its timing requirements.
4. Power Verification: Ensures that the design meets its power consumption requirements.

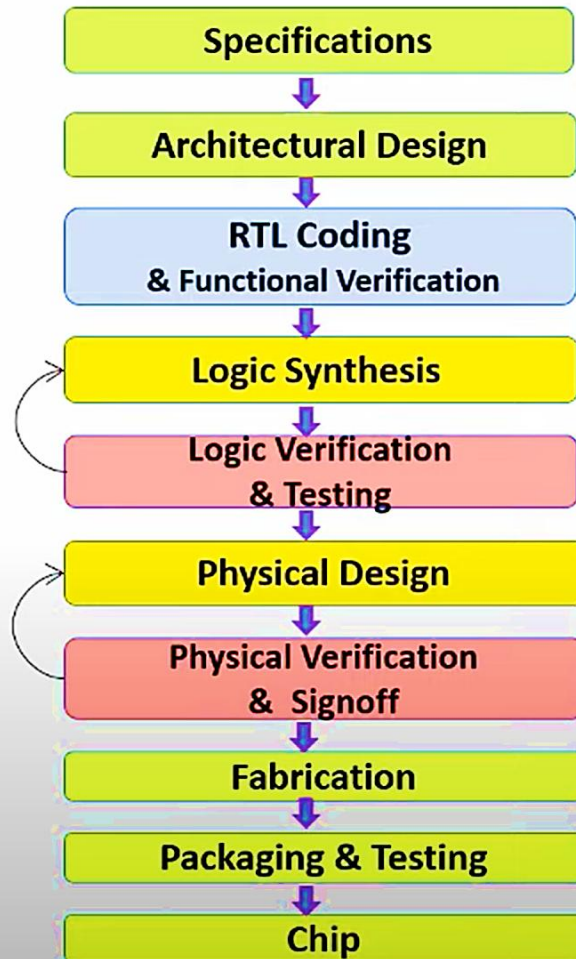


Mixed Signal IC Design Flow

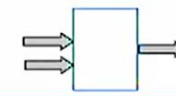


Digital System Design Flow / Digital IC Design Flow ??

VLSI-IC Design Flow / ASIC Flow / RTL to GDS Flow

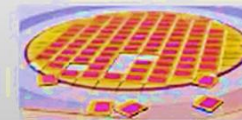
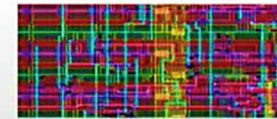
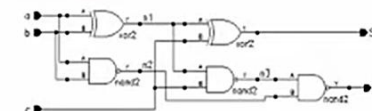


Requirements in terms of Functionality, Power, Performance, Area

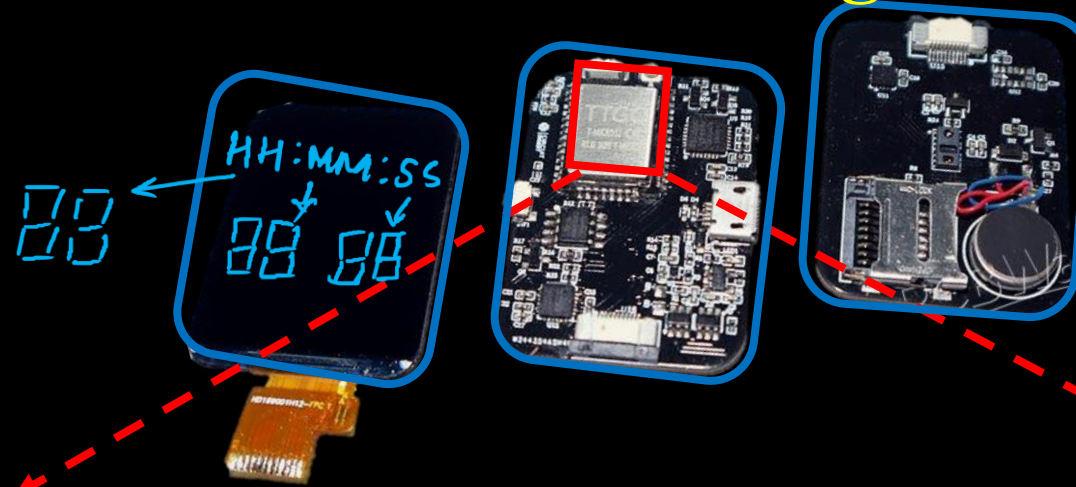


```
always@(A or B or CI)
begin
  S = A ^ B ^ CI;
  CO = (A & B) | (A & CI) | (B & CI);
end
```

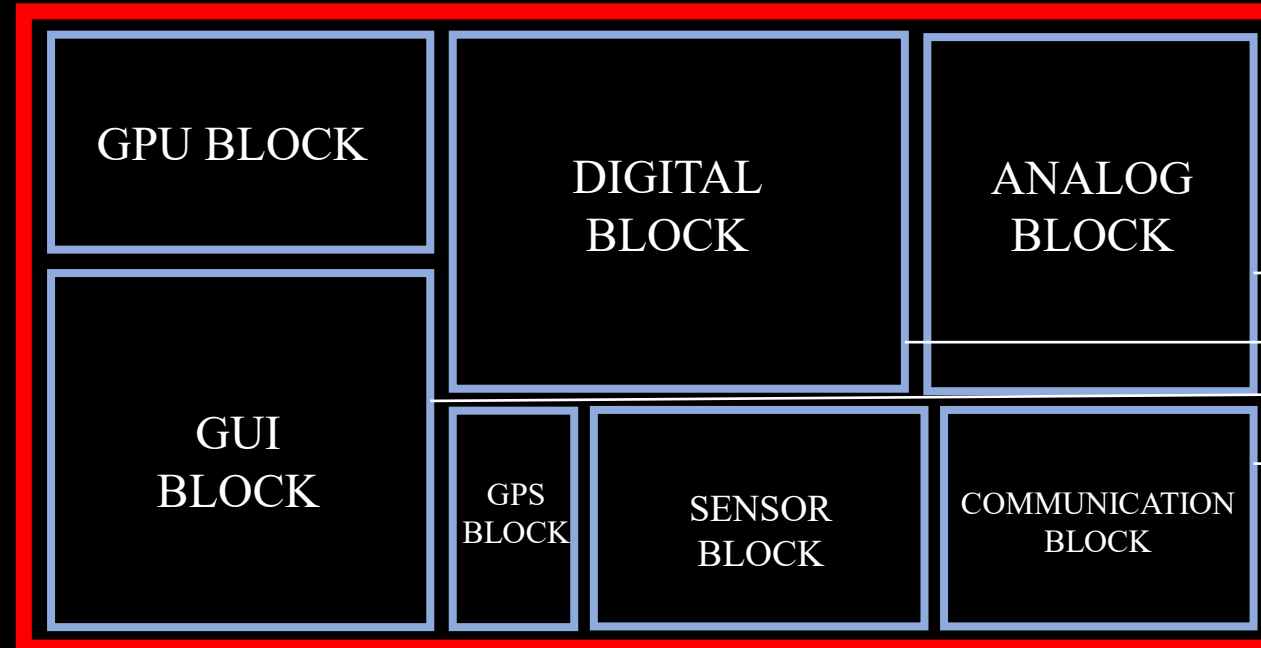
```
module add [co, s, a]
input a, b, c;
output co, s;
xor(n1, a, b);
xor(s, n1, c);
nand(n2, a, b);
nand(n3, n1, c);
nand(co, n3, n2);
endmodule
```



A Real Life Illustration : Digital Smartwatch



ARCHITECTURAL
DESIGN

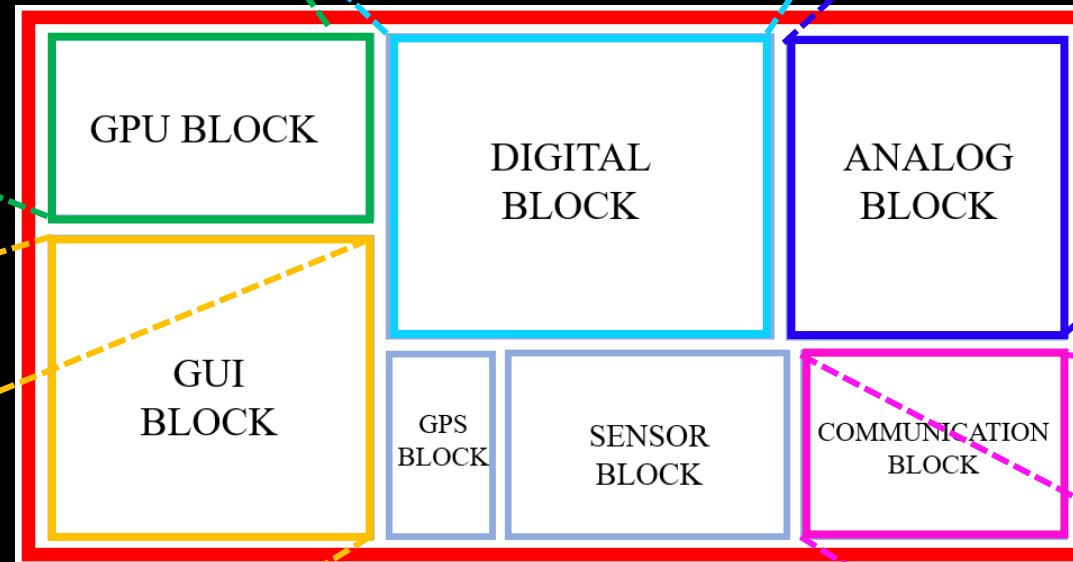


SUB
BLOCKS

- I. Geometry Processor
- II. Rasterizer
- III. Texture Mapping Unit (TMU)
- IV. Shader Core
- V. Framebuffer Controller
- VI. Display Interface
- VII. 2D/3D Acceleration Engine
- VIII. Video Decoder
- IX. Command Processor
- X. Memory Interface (DRAM Controller)

- I. System Clock Generator
- II. Power Management Controller
- III. Task Scheduler
- IV. Interrupt Controller
- V. Real-Time Clock (RTC)
- VI. Boot ROM
- VII. Internal SRAM
- VIII. Cache Controller
- IX. Memory Protection Unit (MPU)
- X. Debug Interface (JTAG/SWD)

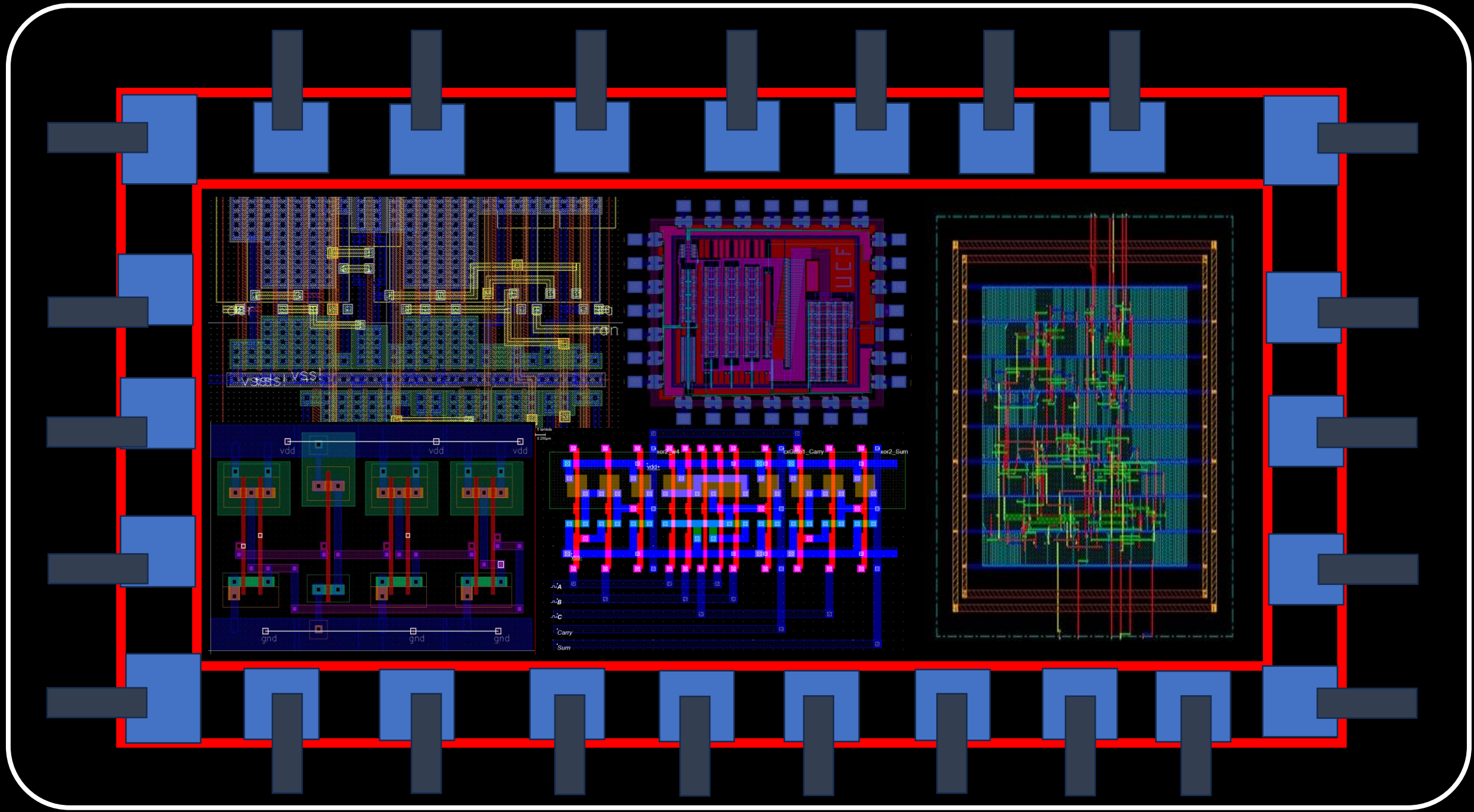
- I. ADC (Analog-to-Digital Converter)
- II. DAC (Digital-to-Analog Converter)
- III. Low-Dropout Regulators (LDOs)
- IV. Oscillator (RC/Crystal)
- V. Bandgap Reference
- VI. Temperature Sensor
- VII. Analog MUX
- VIII. Charge Pump
- IX. Voltage Comparator
- X. Audio Codec (ADC/DAC Pair)



FUNCTIONAL DESIGN

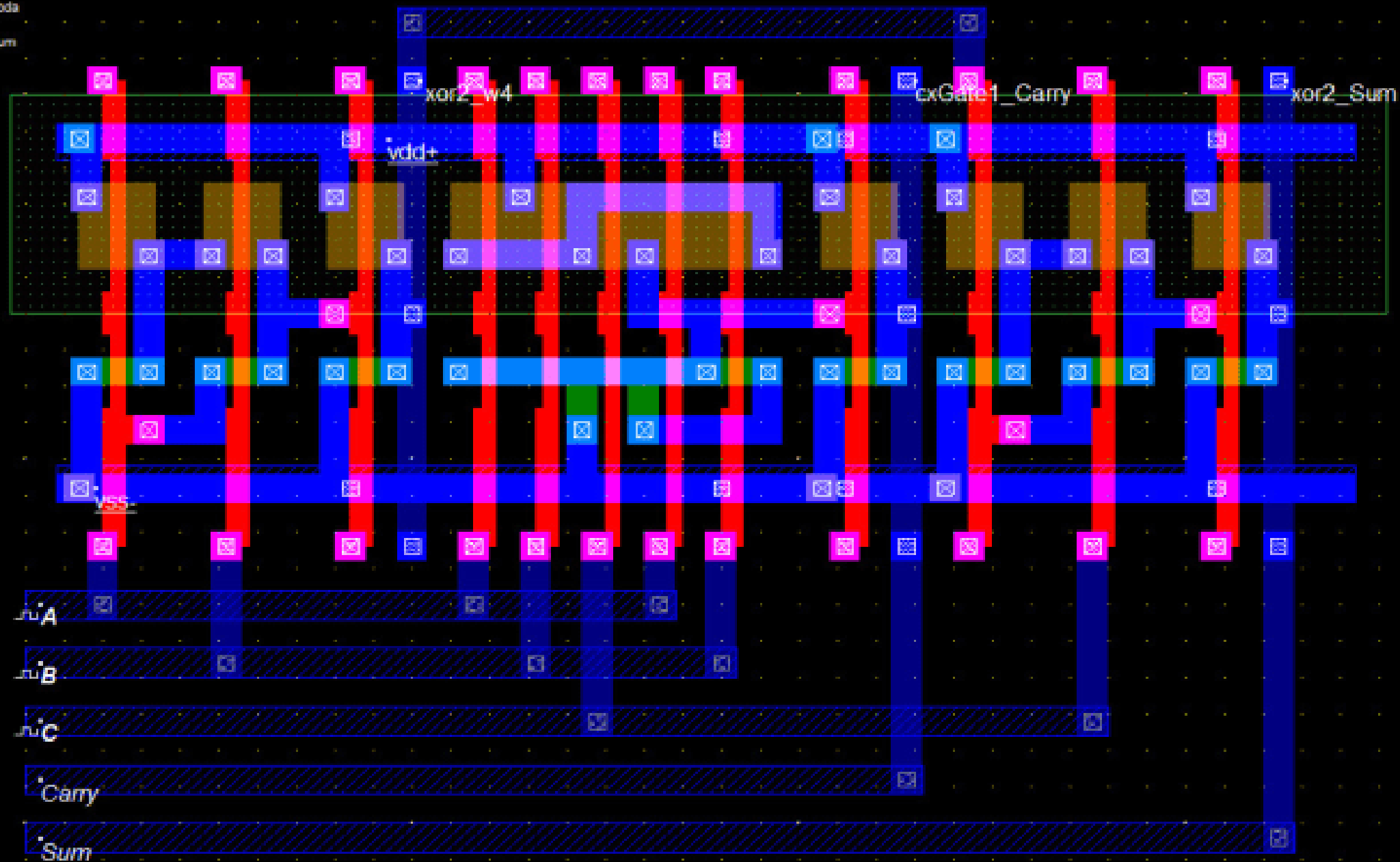
- I. Display Buffer
- II. UI Event Handler
- III. Touch Controller Interface
- IV. Font/Bitmap Engine
- V. Animation Engine
- VI. Frame Scheduler
- VII. Interrupt Controller
- VIII. DMA Engine
- IX. Color Converter
- X. Display Refresh Controller

- I. Bluetooth Transceiver
- II. Wi-Fi Transceiver
- III. LTE Modem
- IV. RF Front-End
- V. Baseband Processor
- VI. MAC Layer Controller
- VII. Antenna Switch Module
- VIII. Clock Synthesizer
- IX. Power Amplifier
- X. Low Noise Amplifier (LNA)



5 lambda

0.250um



digital Watch $\rightarrow$ Counter $\rightarrow$

0 $\rightarrow$ 359

0 $\rightarrow$ 59' = 60 sec $\rightarrow$ 1 min

11 $\leftarrow$ $\begin{matrix} \text{Th} \\ \uparrow \\ 1,00,000 \end{matrix}$ $\rightarrow$ Th
 =
 Hand Ten One
 (MSB) $\rightarrow$ (LSB)

CS 0' ✓
 1' X

	$\downarrow$ A	$\downarrow$ B	Y
0	0	0	
1	0	1	
2	1	0	
3	1	1	

0 $\rightarrow$ 59

64

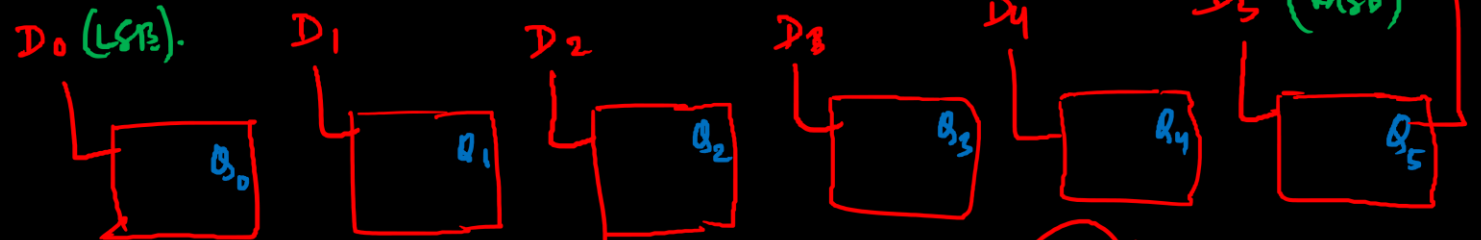
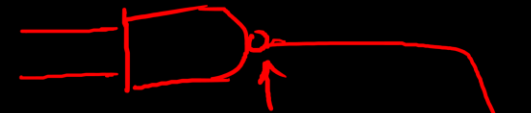
Q = 64 ✓

(6) $\rightarrow$ No. of ffs

0,00,001.

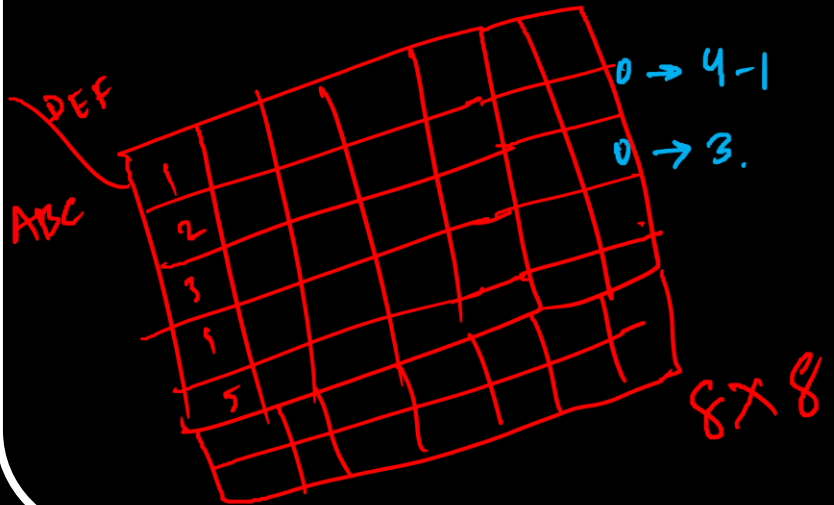
0 ✓

reset - n



2^5	2^4	2^3	2^2	2^1	2^0	decimal
D ₅	D ₄	D ₃	D ₂	D ₁	D ₀	
0	0	0	0	0	0	0

60, 61, 62 } d.c X
 63



0 $\rightarrow$ 2-1
 0 $\rightarrow$ 4-1
 0 $\rightarrow$ 3.